Types

1] row matrix

→ [1 2] 1×2

→ [1 2 3] 1×3

→ [1 2 3 4] 1×4

2] column

→ [1 2]ᵀ = [1; 2] 2×1

[1; 2; 3] 3×1

[1; 2; 3; 4] 4×1

3] rectangular

→ [1 2 3; 4 5 6] 2×3

→ [1 4; 2 5; 3 6] 3×2

4] Zero matrix

[0 0; 0 0]

[0 0 0; 0 0 0; 0 0 0]

5] Square matrix

→ [1 0; 2 4]

[1 3 4; 0 5 -1; 3 6 -1]

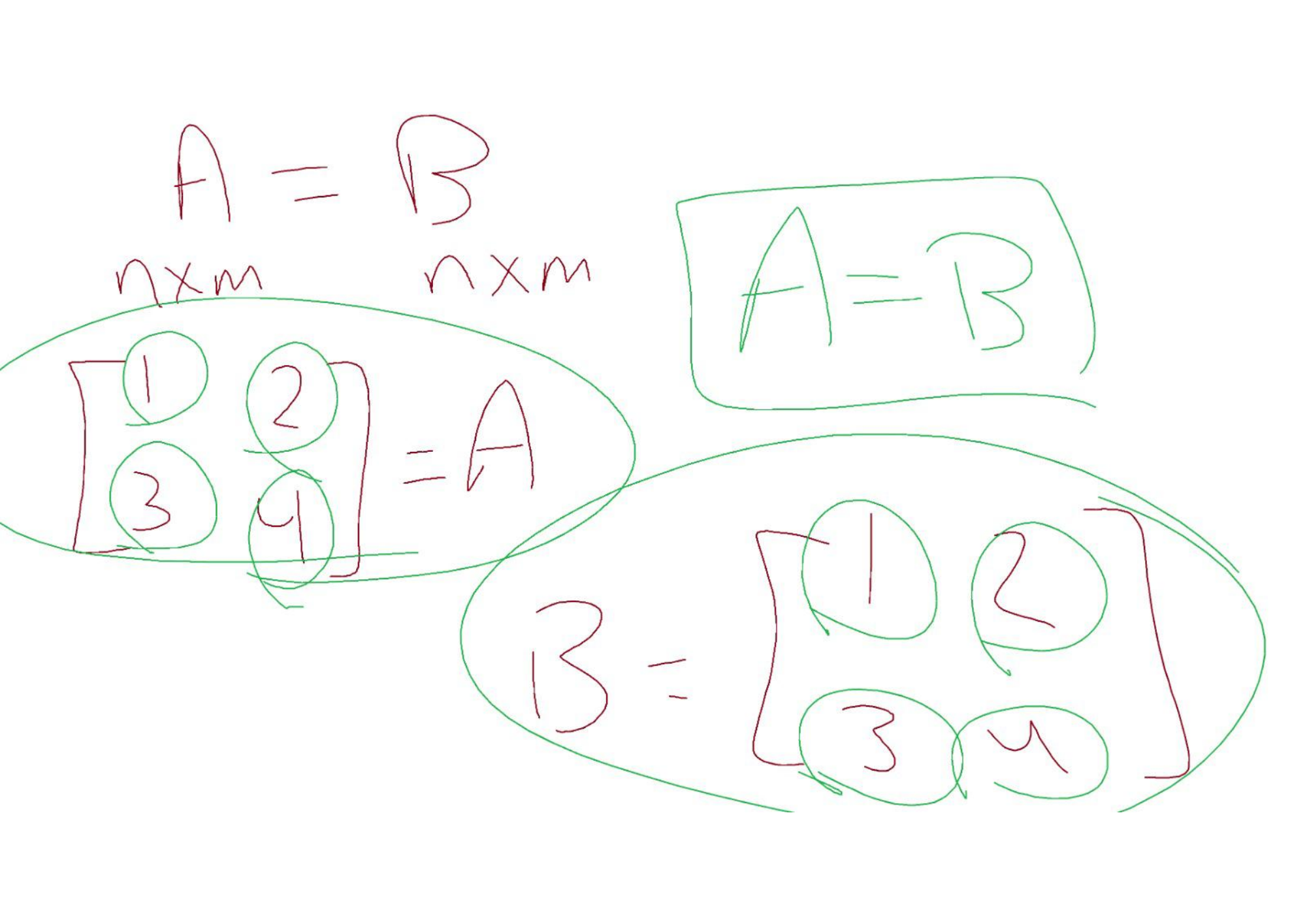
6] identity matrix

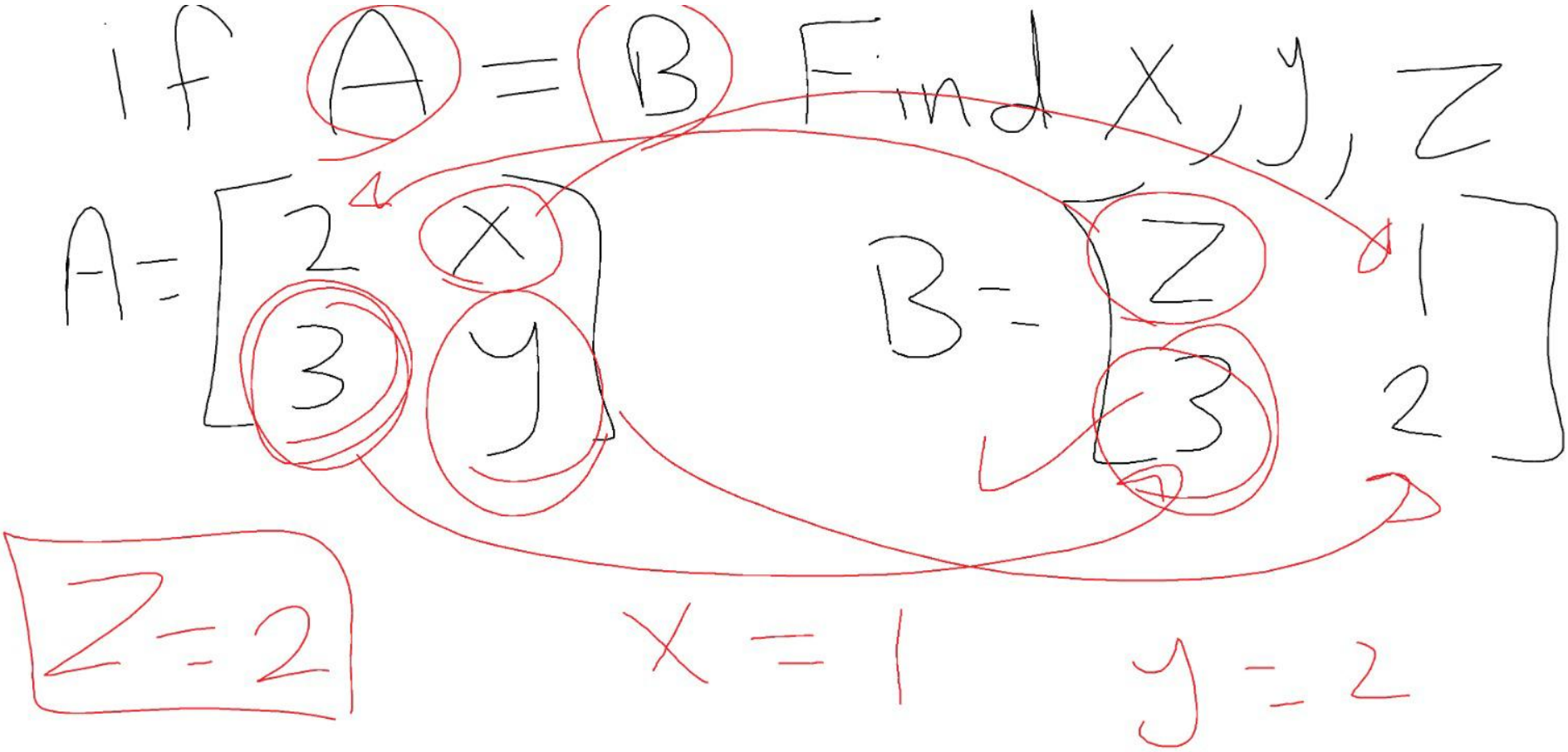
[1 0 0; 0 1 0; 0 0 1]

[1 0; 0 1]

7] Diagonal

[4 0 0; 0 5 0; 0 0 -1]


A = B
n x m
n x m
A = B
1
2
3
4
= A
B =
1
2
3
4


if A = B Find x, y, z
A = [2 x 1]
[3 y 2]
B = [z 1]
[3 2]
z = 2
x = 1
y = 2

$A = B$ Find x, y, z

$$A = \begin{bmatrix} x+y & 3 \\ 4 & x-y \end{bmatrix} \qquad B = \begin{bmatrix} 4 & 3 \\ z & 6 \end{bmatrix}$$

$z = 4$ ✓

$x + y = 4$

$x - y = 6$

$2x = 10 \rightarrow \boxed{x = 5}$

$x + y = 4$

$5 + y = 4$

$\boxed{y = 4 - 5 = -1}$

$$kA = \begin{bmatrix} k() & k() \\ k() & k() \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ -1 & 0 & 5 \end{bmatrix}$$

$$5A = \begin{bmatrix} 10 & 15 & 20 \\ -5 & 0 & 25 \end{bmatrix}$$

transpose matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 3 & 5 & 6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & 3 \\ -1 & 5 \\ 0 & 6 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 4 \\ -1 & 0 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 2 & -1 \\ 4 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 4 & 0 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$$

$A \pm B$

$n \times m$ $n \times m$

No Sol.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2\times 2} \qquad B = \begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix}_{2\times 2}$$

$$A+B = \begin{bmatrix} 6 & 1 \\ 5 & 4 \end{bmatrix}$$

$$B-A = \begin{bmatrix} 4 & -3 \\ -1 & -4 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 0 & 5 \\ 3 & 1 & 6 \end{bmatrix} \quad 2 \times 3$$

$$B = \begin{bmatrix} 2 & 3 \\ -1 & 0 \\ 3 & 5 \end{bmatrix}$$

$A + B =$ no sol

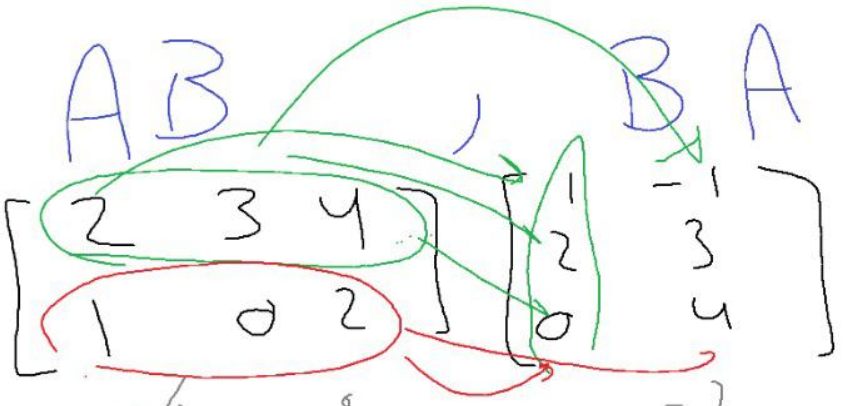
$$B^T = \begin{bmatrix} 2 & -1 & 3 \\ 3 & 0 & 5 \end{bmatrix} \quad 3 \times 2$$

$$A + B^T = \begin{bmatrix} 6 & -1 & 8 \\ 6 & 1 & 11 \end{bmatrix} \quad 2 \times 3$$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 4 \end{bmatrix}$$

Find AB $\quad$ BA

$A \times B =$



$$= \begin{bmatrix} 2\times1+3\times2+4\times0 & 2\times(-1)+3\times3+4\times4 \\ 1\times1+0\times2+2\times0 & 1\times(-1)+0\times3+2\times4 \end{bmatrix} = \begin{bmatrix} 8 & 23 \\ 1 & 7 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 4 \end{bmatrix}$$

Find $AB \neq BA$

$$BA = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 2 \\ 7 & 6 & 15 \\ 4 & 0 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 1\times2 + (-1)\times1 & 1\times3 + (-1)\times0 & 1\times4 + (-1)\times2 \\ 2\times2 + 3\times1 & 2\times3 + 3\times0 & 2\times4 + 3\times2 \\ 0\times2 + 4\times1 & 0\times3 + 4\times0 & 0\times4 + 4\times2 \end{bmatrix}$$